

### 3 Major Changes Needed in LexScan

1. ADD: Physical Product Camera Scanner (Phone-based)

**The 2026 PS says "scanning products, images and labels" — this is NOT just web-crawled images anymore. The system must work when a Legal Metrology officer or consumer points their phone camera at a physical product.**

OLD (LexScan 2025):

Web Crawl → Download Images → OCR → Rule Engine → Dashboard

NEW (LexScan 2026):

TWO MODES:

MODE 1:  LIVE SCAN (Physical Products)

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Phone Camera → Capture Label Image → On-device/Cloud OCR  
→ Rule Engine → Instant Compliance Result on Screen

MODE 2:  BATCH SCAN (E-Commerce - still valid, just secondary)

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Web Crawl → Download Images → OCR → Rule Engine → Dashboard

#### **Implementation:**

- Add a **React-based PWA / mobile-responsive web app** with camera access (`navigator.mediaDevices.getUserMedia`)
- Officer opens app → taps "Scan Product" → camera activates → captures label → sends to FastAPI backend → gets instant compliance verdict
- This becomes your **primary demo mode** — way more impressive than web crawling

#### **IMPORTANT**

**This is actually a huge upgrade for your demo. Instead of showing a crawled webpage, a judge can literally hand you any product (water bottle, chips packet, pen) and you scan it live on stage. Showstopper moment.**

2. ADD: Font Size Validation (New CV Feature)

**The PS explicitly calls out "incorrect font sizes" as a violation. Under the Legal Metrology Rules, declarations must meet minimum font size requirements based on package area:**

| Package Surface Area       | Minimum Font Height (Numeral) |
|----------------------------|-------------------------------|
| Up to 100 cm <sup>2</sup>  | 1 mm                          |
| 100–500 cm <sup>2</sup>    | 2 mm                          |
| 500–2500 cm <sup>2</sup>   | 4 mm                          |
| Above 2500 cm <sup>2</sup> | 6 mm                          |

### New CV Pipeline Addition:

After OCR extracts text regions:



[Font Size Estimator]

- └── Use OpenCV contour analysis on detected text bounding boxes
- └── Measure pixel height of characters
- └── Estimate real-world mm using reference object or known package

dimensions

- └── Compare against minimum font size table
- └── Flag: "MRP font size = ~1.5mm on a 600cm<sup>2</sup> package.  
Minimum required = 4mm. VIOLATION."

**This is a technically impressive differentiator — most competing teams will only check IF text exists, not whether the font size is legally compliant.**

### 3. SHIFT: Format & Layout Compliance (Not Just Content)

**The PS says "prescribed declarations in a specified format and manner". This means:**

- MRP must be printed as **"MRP ₹XXX (inclusive of all taxes)"** — not just "₹XXX" or "Price: XXX"
- Net quantity must use **standard units** — "500 g" not "500 gm" or "500 grams"
- Manufacturer address must include **full postal address with PIN code**
- "Best before" must follow specific date format patterns

### Rule Engine Enhancement:

python

# OLD: Just check if MRP exists

```
rules = {
    "mrp": lambda text: bool(re.search(r'₹\d+', text)) # Too basic
}
```

# NEW: Check format compliance too

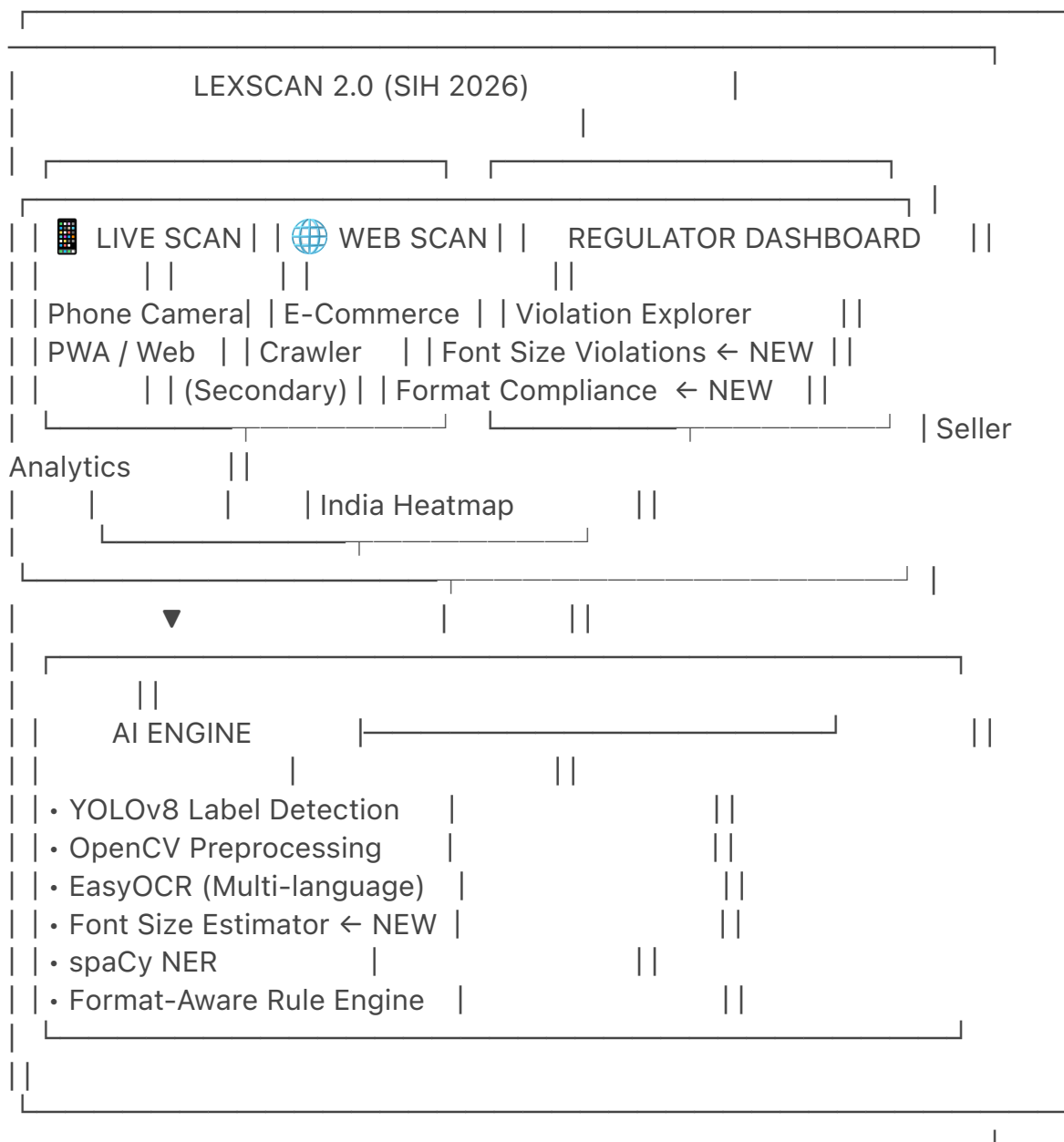
```
rules = {
    "mrp_present": lambda text: bool(re.search(r'₹\d+', text)),
    "mrp_format": lambda text: bool(re.search(
        r'M\.?R\.?P\.?s*[\₹:]?s*\d+.*inclusive.*tax', text, re.I
    ))
}
```

```

)), # Must say "inclusive of all taxes"
"net_qty_unit": lambda text: validate_standard_units(text),
# "500 g" ✓, "500 gm" ✗, "500 grams" ✗
"mfg_date_format": lambda text: bool(re.search(
    r'(Mfg|Mfd|Manufactured).*\d{2}/\d{4}', text, re.I
)),
}

```

## Updated LexScan Architecture (2026 Version)



## What DOESN'T Change (Still Valid)

| Component | Status |
|-----------|--------|
|-----------|--------|

|                                 |                                       |
|---------------------------------|---------------------------------------|
| FastAPI backend                 | ✓ Same                                |
| PostgreSQL + Redis              | ✓ Same                                |
| Next.js Dashboard (all 5 pages) | ✓ Same                                |
| EasyOCR + OpenCV pipeline       | ✓ Same (enhanced)                     |
| YOLOv8 label segmentation       | ✓ Same                                |
| spaCy NER extraction            | ✓ Same                                |
| Celery task queue               | ✓ Same                                |
| ₹0 free-tier cost model         | ✓ Same                                |
| Team of 5 task split            | ✓ Same (minor role adjustments)       |
| Judge Q&A prep                  | ✓ Mostly same (add font size answers) |